

Knowledge and Data

Working Group

Week 2: RDF & SPARQL



Question 1—Modelling tabular information as a graph

Consider the following *relational database* tables:

Students

<i>student</i>	<i>VUnetID</i>	<i>uni</i>	<i>study</i>
s1	as344	UvA	IK
s2	ex444	VU	IMM

StudyProgrammes

<i>study</i>	<i>name</i>	<i>language</i>
IMM	Information, Management, and Multimedia	Dutch
IK	Informatiekunde	Dutch

Task: Construct a knowledge graph G that represents this knowledge, using our simple triple notation:

(*subject*, *predicate*, *object*)

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(s1, a, Student)
(s2, a, Student)

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(s1, hasVUNetID, "as344")
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(s1, studiesAt, UvA)
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(s1, followsStudy, IK)
(s2, followsStudy, IMM)

Question 1—Modelling tabular information as a graph

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Question 1—Modelling tabular information as a graph

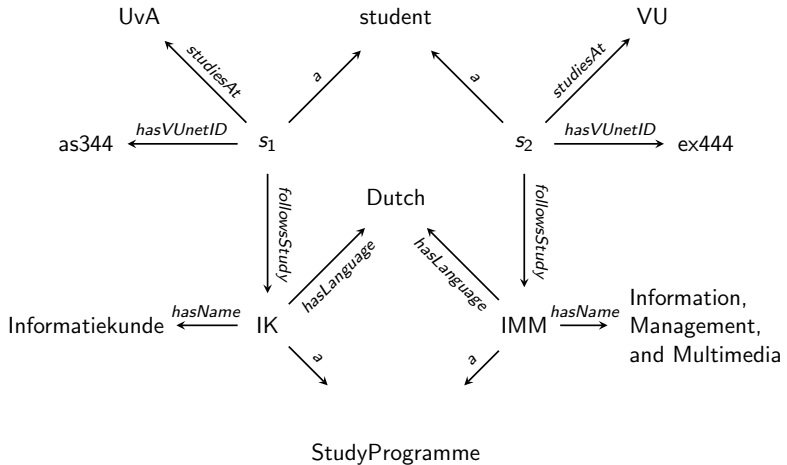
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(IMM, hasLanguage, Dutch)
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Question 1—Modelling tabular information as a graph

Visualized as a graph:



Question 2—RDF and Turtle

Consider once more our knowledge graph G .

Task: Serialize the triples that make up G in *Turtle* syntax. Such that you:

1. make use of namespaces and URIs / IRIs,
2. abbreviate namespaces using prefixes, and
3. use one or more syntactic shortcuts.

Question 2—RDF and Turtle

The entire graph:

(s1,	a,	Student)
(s2,	a,	Student)
(s1,	hasVUnetID,	"as344")
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(s1,	studiesAt,	UvA)
(s2,	studiesAt,	VU)
(s1,	followsStudy,	IK)
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(IMM,	a,	StudyProgramme)
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(IMM,	hasLanguage,	Dutch)
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Turtle specification: <https://www.w3.org/TR/turtle>

Question 2—RDF and Turtle

```
1 @prefix ex: <http://www.example.org/> .
2 @prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
3 @prefix dbr: <http://dbpedia.org/resource/> .
4 @prefix dbo: <http://dbpedia.org/ontology/> .
5 @prefix xsd: <http://www.w3.org/2001/XMLSchema#> .
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12 ex:s2 a dbr:Student ; # shorthand syntax
13     ex:hasVUNetID "ex444"^^xsd:string ; # datatype annotation
14     ex:studiesAt ex:VU ;
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```

Question 2—RDF and Turtle

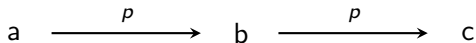
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16
17 ex:IK a ex:StudyProgramme ;
18     ex:hasName "Informatiekunde"@nl ; # language tag
19     ex:hasLanguage dbr:Dutch .
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```

Question 3—Models for knowledge graphs

Consider a knowledge graph H with triples (a, p, b) and (b, p, c) :



Let an interpretation I consist of universe $U = \{1, 2\}$

Which of the following interpretations is a model of H :

1. $I(a) = 2, I(b) = 1, I(c) = 1, I(p) = \{(1, 2)\}$
2. $I(a) = 2, I(b) = 1, I(c) = 2, I(p) = \{(1, 2)\}$
3. $I(a) = 2, I(b) = 1, I(c) = 2, I(p) = \{(2, 2)\}$
4. $I(a) = 2, I(b) = 1, I(c) = 1, I(p) = \{(2, 2), (1, 1)\}$
5. $I(a) = 2, I(b) = 2, I(c) = 2, I(p) = \{(2, 2)\}$

Remember: I is a model of a knowledge graph iff it is a model of *all* its triples, and I is a model of a triple iff the interpretation is true

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5. $I(a) = 2, I(b) = 2, I(c) = 2, I(p) = \{(2, 2)\}$
 $(2, p, 2); (2, p, 2); I(p)$ holds for (2, 2)

Question 4—Entailments for knowledge graphs

Let K be a knowledge graph defined by triples

(Victor,	teaches,	KaD)
(Stefan,	a,	Teacher)
(KaD,	a,	Course)
(Carl,	participatesIn,	KaD)
(Carl,	a,	Student)

Which of the following knowledge graphs is entailed by K ?

- | | |
|--|--|
| 1) (Carl, participatesIn, Course) | 3) (Stefan, a, Teacher)
(Carl, a, Student)
(KaD, a, Course) |
| 2) (Victor, teaches, Course)
(Stefan, a, Teacher)
(KaD, a, Course)
(Carl, participatesIn, Course) | 4) (Victor, teaches, KaD)
(Stefan, a, Teacher)
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(Carl, a, Student)
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Question 5—Resources

Describe, in your own words, the difference between a URI/IRI and a Literal

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- non-tangible objects, like concepts and dreams
- relationships between resources

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URIs/IRIs uniquely identify

- tangible objects (“things”), like people, cars, and molecules
- non-tangible objects, like concepts and dreams
- relationships between resources

Literals are attribute-values of (non-) tangible objects

Question 6—Exploring DBpedia

Consider the following two DBpedia resources:

- a) <https://dbpedia.org/resource/Amsterdam>
- b) https://dbpedia.org/resource/Kingdom_of_the_Netherlands

Which of the follow three triples together form a path from **a** to **b**:

1. (dbr:Netherlands, dbo:leaderName, dbr:Willem-Alexander_of_the_Netherlands)
2. (dbr:Kingdom_of_the_Netherlands, dbo:language, dbr:Dutch_language)
3. (dbr:Amsterdam, rdf:type, dbo:Place)
4. (dbr:Kingdom_of_the_Netherlands, dbo:leaderName, dbr:Willem-Alexander_of_the_Netherlands)
5. (dbr:Netherlands, dbo:part_of, dbr:Kingdom_of_the_Netherlands)
6. (dbr:Willem-Alexander_of_the_Netherlands, rdf:type, dbo:Person)
7. (dbr:Amsterdam, dbo:country, dbr:Netherlands)

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- 3. (**dbr:Amsterdam**, rdf:type, dbo:Place)
- 4. (dbr:Kingdom_of_the_Netherlands, dbo:leaderName, dbr:Willem-Alexander_of_the_Netherlands)
- 5. (dbr:Netherlands, dbo:part_of, dbr:Kingdom_of_the_Netherlands)
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Which of the follow three triples together form a path from **a** to **b**:

- 1. (dbr:Netherlands, dbo:leaderName, dbr:Willem-Alexander_of_the_Netherlands)
- 2. (dbr:Kingdom_of_the_Netherlands, dbo:language, dbr:Dutch_language)
- 3. (dbr:Amsterdam, rdf:type, dbo:Place)
- 4. (**dbr:Kingdom_of_the_Netherlands**, dbo:leaderName, dbr:Willem-Alexander_of_the_Netherlands)
- 5. (dbr:Netherlands, dbo:part_of, dbr:Kingdom_of_the_Netherlands)
- 6. (dbr:Willem-Alexander_of_the_Netherlands, rdf:type, dbo:Person)
- 7. (dbr:Amsterdam, dbo:country, dbr:Netherlands)

Question 7—DBpedia & SPARQL

Consider following SPARQL query:

```
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX dbo: <http://dbpedia.org/ontology/>
SELECT ?capital ?area
WHERE {
    ?country dbo:capital ?capital.
    ?capital dbo:areaTotal ?area .
}
```

Execute this query against the DBpedia SPARQL endpoint [1] in YASGUI [2].

[1] <http://dbpedia.org/sparql>

[2] <https://yasgui.triply.cc>

Question 7—DBpedia & SPARQL

Query ✕ +

https://dbpedia.org/sparql

```
1 PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
2 PREFIX dbo: <http://dbpedia.org/ontology/>
3 SELECT ?capital ?area
4 WHERE {
5   ?country dbo:capital ?capital.
6   ?capital dbo:areaTotal ?area .
7 }
```

Table Response Gallery Chart Geo Geo-3D Geo events Pivot Timeline 3757 results

capital	area
1 <http://dbpedia.org/resource/Balta_Odessa_Oblast>	"2.297e+07"^^<http://www.w3.org/2001/XMLSchema#double>
2 <http://dbpedia.org/resource/Bamako>	"2.45e+08"^^<http://www.w3.org/2001/XMLSchema#double>
3 <http://dbpedia.org/resource/Bamako>	"2.45e+08"^^<http://www.w3.org/2001/XMLSchema#double>
4 <http://dbpedia.org/resource/Bamako>	"2.45e+08"^^<http://www.w3.org/2001/XMLSchema#double>
5 <http://dbpedia.org/resource/Bamberg>	"5.462e+07"^^<http://www.w3.org/2001/XMLSchema#double>
6 <http://dbpedia.org/resource/Bamberg>	"5.462e+07"^^<http://www.w3.org/2001/XMLSchema#double>
7 <http://dbpedia.org/resource/Bamyan>	"3.5e+07"^^<http://www.w3.org/2001/XMLSchema#double>
8 <http://dbpedia.org/resource/Banaba>	"6.29e+06"^^<http://www.w3.org/2001/XMLSchema#double>

3 884 results*

* on 26 June 2022

Question 8—Eliminate duplicates with SPARQL

Consider, once again, the following SPARQL query:

```
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX dbo: <http://dbpedia.org/ontology/>
SELECT ?capital ?area
WHERE {
    ?country dbo:capital ?capital.
    ?capital dbo:areaTotal ?area .
}
```

Adjust the query to eliminate duplicate results. How many remain?

SPARQL specification: <https://www.w3.org/TR/sparql11-query/>

Question 8—Eliminate duplicates with SPARQL

Query ✕ +

https://dbpedia.org/sparql

```
1 PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
2 PREFIX dbo: <http://dbpedia.org/ontology/>
3 SELECT DISTINCT ?capital ?area
4 WHERE {
5   ?country dbo:capital ?capital.
6   ?capital dbo:areaTotal ?area .
7 }
```

Table Response Gallery Chart Geo Geo-3D Geo events Pivot Timeline 1880 results

capital	area
<http://dbpedia.org/resource/Balta_Odessa_Oblast>	"2.297e+07"^^<http://www.w3.org/2001/XMLSchema#double>
<http://dbpedia.org/resource/Bamako>	"2.45e+08"^^<http://www.w3.org/2001/XMLSchema#double>
<http://dbpedia.org/resource/Bamberg>	"5.462e+07"^^<http://www.w3.org/2001/XMLSchema#double>
<http://dbpedia.org/resource/Bamyan>	"3.5e+07"^^<http://www.w3.org/2001/XMLSchema#double>
<http://dbpedia.org/resource/Banaba>	"6.29e+06"^^<http://www.w3.org/2001/XMLSchema#double>
<http://dbpedia.org/resource/Banda_Aceh>	"6.136e+07"^^<http://www.w3.org/2001/XMLSchema#double>
<http://dbpedia.org/resource/Bandar_Seri_Begawan>	"1.0036e+08"^^<http://www.w3.org/2001/XMLSchema#double>
<http://dbpedia.org/resource/Bandung>	"1.6731e+08"^^<http://www.w3.org/2001/XMLSchema#double>

1 936 results remain after adding the **DISTINCT** keyword*

Question 9—SPARQL Advanced

Consider our improved SPARQL query:

```
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX dbo: <http://dbpedia.org/ontology/>
SELECT DISTINCT ?capital ?area
WHERE {
    ?country dbo:capital ?capital.
    ?capital dbo:areaTotal ?area .
}
```

Restrict the query to include only bindings to ?capital that have a *rdfs:label* relation with the English language literals “Amsterdam” or “Berlin”.

Note that there is more than one solution

Question 9—SPARQL Advanced

Solution A: add a **filter**

```
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX dbpedia: <http://dbpedia.org/resource/>
SELECT DISTINCT ?capital ?area
WHERE {
    ?country dbo:capital ?capital .
    ?capital dbo:areaTotal ?area ;
              rdfs:label ?label .

    FILTER (?label = "Amsterdam"@en || ?label = "Berlin"@en)
}
```

	capital	 area
1	dbpedia:Amsterdam	"2.1932e+08"^^<http://www.w3.org/2001/XMLSchema#double>
2	dbpedia:Berlin	"8.917e+08"^^<http://www.w3.org/2001/XMLSchema#double>

Showing 1 to 2 of 2 entries

Question 9—SPARQL Advanced

Solution B: perform a **union**

```
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX dbpedia: <http://dbpedia.org/resource/>
SELECT DISTINCT ?capital ?area
WHERE {
    ?country dbo:capital ?capital .
    ?capital dbo:areaTotal ?area .

    { ?capital rdfs:label "Amsterdam"@en }
    UNION
    { ?capital rdfs:label "Berlin"@en }
}
```

	capital	 area
1	dbpedia:Amsterdam	"2.1932e+08"^^< http://www.w3.org/2001/XMLSchema#double >
2	dbpedia:Berlin	"8.917e+08"^^< http://www.w3.org/2001/XMLSchema#double >

Showing 1 to 2 of 2 entries

Question 10—SPARQL Advanced II

Consider our filtered SPARQL query:

```
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX dbpedia: <http://dbpedia.org/resource/>
SELECT DISTINCT ?capital ?area
WHERE {
    ?country dbo:capital ?capital .
    ?capital dbo:areaTotal ?area ;
              rdfs:label ?label .

    FILTER (?label = "Amsterdam"@en || ?label = "Berlin"@en)
}
```

Extend the query to also retrieve the value of the *dbo:areaWater* property and bind it to a variable *?areaWater*.

Make sure that it still returns both Amsterdam and Berlin.

Question 10—SPARQL Advanced II

Naive solution:

```
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX dbpedia: <http://dbpedia.org/resource/>
SELECT DISTINCT ?capital ?area ?areaWater
WHERE {
    ?country dbo:capital ?capital .
    ?capital dbo:areaTotal ?area ;
             dbo:areaWater ?areaWater ;
             rdfs:label ?label .

    FILTER (?label = "Amsterdam"@en || ?label = "Berlin"@en)
}
```

	capital	area	areaWater
1	dbpedia:Amsterdam	"2.1932e+08"^^<http://www.w3.org/2001/XMLSchema#double>	"5.356e+07"^^<http://

Showing 1 to 1 of 1 entries

Question 10—SPARQL Advanced II

Naive solution:

```
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX dbpedia: <http://dbpedia.org/resource/>
SELECT DISTINCT ?capital ?area ?areaWater
WHERE {
    ?country dbo:capital ?capital .
    ?capital dbo:areaTotal ?area ;
              dbo:areaWater ?areaWater ;
              rdfs:label ?label .

    FILTER (?label = "Amsterdam"@en || ?label = "Berlin"@en)
}
```

	capital	area	areaWater
1	dbpedia:Amsterdam	"2.1932e+08"^^<http://www.w3.org/2001/XMLSchema#double>	"5.356e+07"^^<http://

Showing 1 to 1 of 1 entries

We are missing Berlin!

Question 10—SPARQL Advanced II

Let us inspect the resource Berlin:



```
1 PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
2 PREFIX dbo: <http://dbpedia.org/ontology/>
3 PREFIX dbpedia: <http://dbpedia.org/resource/>
4 SELECT DISTINCT ?capital ?p ?o
5 WHERE {
6     ?country dbo:capital ?capital .
7     ?capital rdfs:label "Berlin"@en ;
8             ?p ?o .
9
10    FILTER regex(?p, "area")
11 }
```

 Table

 Response

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5 res

	capital	p	o
1	dbpedia:Berlin	<http://dbpedia.org/property/areaCode>	"30"^^<http://www.w3.org/2001/XMLSchema#integer>
2	dbpedia:Berlin	<http://dbpedia.org/property/areaTotalKm>	"891.7"^^<http://www.w3.org/2001/XMLSchema#double>
3	dbpedia:Berlin	dbo:areaCode	030
4	dbpedia:Berlin	dbo:areaTotal	"8.917e+08"^^<http://www.w3.org/2001/XMLSchema#double>
5	dbpedia:Berlin	dbo:PopulatedPlace/areaTotal	"891.7"^^<http://dbpedia.org/datatype/squareKilometre>

Berlin does not contain any triples that use the *dbo:areaWater* predicate

Question 10—SPARQL Advanced II

Solution: make the requirement **optional**

```
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX dbpedia: <http://dbpedia.org/resource/>
SELECT DISTINCT ?capital ?area ?areaWater
WHERE {
    ?country dbo:capital ?capital .
    ?capital dbo:areaTotal ?area ;
        rdfs:label ?label .

    OPTIONAL { ?capital dbo:areaWater ?areaWater . }
    FILTER (?label = "Amsterdam"@en || ?label = "Berlin"@en)
}
```

	capital	area	areaWater
1	dbpedia:Amsterdam	"2.1932e+08"^^<http://www.w3.org/2001/XMLSchema#double>	"5.356e+07"^^<http://www.w3.org/2001/XMLSchema#double>
2	dbpedia:Berlin	"8.917e+08"^^<http://www.w3.org/2001/XMLSchema#double>	

Showing 1 to 2 of 2 entries

Quiz

To test your proficiency:

Exercise quiz Module 2
(see the course's Canvas page)

NB: completing the quiz is mandatory to continue to next week's module